

GOSTOMEL, Ukraine

WMO: 333451

Lat: 50.603N

Lon: 30.192E

Elev: 517

StdP: 14.42

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.9	4.9	-7.5	3.7	1.3	-0.8	5.4	6.6	22.1	36.4	20.1	35.8	5.7	40	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	89.9	67.7	86.3	66.7	83.9	65.7	71.1	81.7	69.6	80.5	68.0	78.7	8.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.1	105.1	74.9	66.3	98.6	73.3	64.5	92.7	72.0	35.2	82.4	33.9	81.1	32.7	78.0	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.0	17.5	15.6	DB	-6.8	94.6	7.4	3.7	-12.2	97.3	-16.5	99.5	-20.7	101.6	-26.1	104.3
			WB	-7.2	73.3	7.4	2.5	-12.5	75.1	-16.9	76.6	-21.1	78.0	-26.4	79.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.3	24.8	26.2	36.1	49.9	60.6	67.0	69.8	68.5	59.0	47.3	38.2	30.3
	DBStd	17.96	10.81	10.88	8.74	7.30	7.49	6.02	5.35	6.61	7.07	7.65	7.77	8.76
	HDD50	3102	781	666	435	89	9	0	0	0	9	143	360	610
	HDD65	6590	1246	1086	897	455	176	49	16	37	203	547	803	1075
	CDD50	2476	0	0	3	87	338	510	615	574	281	61	7	0
	CDD65	489	0	0	0	3	40	109	167	146	24	0	0	0
	CDH74	4954	0	0	0	39	445	1037	1543	1610	278	2	0	0
	CDH80	1622	0	0	0	9	98	290	512	651	62	0	0	0
	WSAvg	7.6	8.6	7.9	8.9	7.8	7.2	7.1	6.8	6.5	6.8	7.0	8.1	8.7
	PrecAvg	23.4	1.4	1.4	1.4	1.6	2.4	3.0	2.9	2.3	2.1	1.7	1.7	1.6
Precipitation	PrecMax	29.4	2.2	3.0	4.1	4.4	6.3	6.2	6.2	4.6	7.0	4.3	3.5	5.2
	PrecMin	17.4	0.4	0.4	0.2	0.1	0.8	0.4	1.0	0.2	0.0	0.0	0.1	0.4
	PrecStd	2.8	0.5	0.6	0.9	0.9	1.3	1.5	1.3	1.2	1.8	1.1	0.8	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.6	50.2	62.8	80.3	86.4	89.7	93.2	96.9	86.2	73.3	61.0	51.6
		MCWB	43.2	44.6	50.6	60.1	64.9	67.3	69.8	67.2	64.8	58.2	54.2	48.8
	2%	DB	43.0	44.2	57.3	71.7	82.5	86.2	89.3	91.6	80.6	68.0	55.7	46.8
		MCWB	41.0	40.3	46.6	54.8	62.2	65.8	68.2	67.9	62.0	57.9	51.9	45.0
	5%	DB	40.6	40.8	53.3	67.7	79.0	83.8	85.8	87.5	77.0	62.9	52.0	44.4
		MCWB	38.9	37.9	44.1	53.4	61.4	65.5	68.2	66.3	61.3	54.9	48.6	42.8
	10%	DB	37.4	37.7	48.5	64.1	75.3	80.5	82.4	82.6	73.1	59.0	49.6	41.2
		MCWB	36.4	35.5	42.2	51.4	60.8	64.7	66.3	64.8	59.9	53.3	47.3	40.0
	0.4%	WB	43.9	46.2	50.8	60.5	68.6	71.6	73.9	72.7	67.5	61.2	55.5	49.6
		MCDB	45.3	49.6	60.9	76.8	81.9	80.4	85.0	84.3	82.2	68.2	59.1	50.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.2	41.1	48.0	57.3	65.5	69.5	71.8	70.4	64.8	58.6	52.9	45.5
		MCDB	42.1	43.1	55.9	68.3	76.2	79.8	82.2	82.3	76.2	65.2	56.1	46.9
	5%	WB	38.5	38.1	45.1	54.8	63.6	67.8	70.2	68.6	62.7	56.3	49.6	42.5
		MCDB	39.6	39.8	50.3	63.8	74.3	78.1	80.5	80.4	73.0	61.9	51.6	43.5
	10%	WB	36.0	35.8	42.7	52.7	61.7	66.4	68.5	67.2	60.9	53.8	47.1	39.9
		MCDB	37.0	37.1	47.8	61.7	72.0	76.9	78.1	78.7	70.0	58.8	49.0	41.1
Mean Daily Temperature Range	5% DB	MDBR	8.9	10.3	13.8	18.6	20.0	19.9	20.0	20.8	19.0	15.4	8.8	7.7
		MCDBR	8.7	12.0	21.5	24.3	25.0	24.5	24.4	27.2	25.6	21.7	13.4	10.8
		MCWBR	7.7	9.0	12.7	11.8	9.8	9.1	9.1	8.6	11.0	12.4	10.1	8.9
	5% WB	MCDBR	8.2	10.4	18.1	20.7	21.9	20.9	20.4	22.1	22.0	19.1	11.5	9.5
		MCWBR	7.6	8.2	12.1	10.9	9.8	9.6	8.6	9.0	10.7	12.1	9.5	8.4
Clear-Sky Solar Irradiance	taub	0.307	0.339	0.371	0.433	0.408	0.400	0.429	0.432	0.399	0.360	0.341	0.308	
	taud	2.213	2.158	2.213	2.130	2.284	2.327	2.257	2.237	2.288	2.373	2.308	2.252	
	Ebn at Noon	221	244	260	256	269	272	261	253	248	238	208	202	
	Edh at Noon	23	32	36	44	40	38	41	40	34	26	22	19	
All-Sky Solar Radiation	RadAvg	269	530	939	1371	1784	1886	1822	1586	1085	620	248	182	
	RadStd	47	94	149	141	144	140	123	97	161	100	35	44	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (9 neighbors)	+1.35	N/A	N/A	+1.75	+1.04	+0.97	-252	-364	+211	+94

Nomenclature: See separate page